From Aphids to Algorithms: Why AI Deserves Behavioural Study

Introduction: Behavioural Syndrome ≠ Internal Experience

This Commentary argues that claims about stable behavioural tendencies in large language models are frequently misconstrued as anthropomorphism or implicit assumptions about internal states. Ethological research has long addressed analogous concerns in the study of non-human animals, offering a well-established methodological precedent for making personality-like claims without invoking human psychology or subjective experience.
Large language models exhibit stable behavioural syndromes identical in structure to those studied in parthenogenetic aphids, yet remain exempt from the ethological toolkit we routinely apply to insects. This exemption is inconsistent with existing practice and weakens alignment efforts. Rigorous observation of algorithms is no longer optional.

Behavioural syndromes, suites of correlated behaviors expressed consistently across contexts, have been documented across taxa with vastly different neural architectures, from parthenogenetic aphids to chimpanzees (Schuett *et al.* 2011; Murray,1998). This framework has proven valuable for understanding behavioural trade-offs, ecological implications, and evolutionary constraints without requiring strong assumptions about internal experience. We propose that this same framework should be systematically applied to artificial intelligence systems, not because they are conscious, but because the logical prerequisites for behavioural syndrome analysis are met more readily with AI than with many biological organisms we already study.

Keywords: Anthropomorphosization, artificial intelligence, behavioural syndromes, comparative ethology, computational ethology, personality

The Aphid Problem (Or: Who's Really Anthropomorphizing Here?)

Consider the pea aphid (*Acyrthosiphon pisum*), a clonal insect with approximately 100,000 neurons. Schuett *et al.* (2011) demonstrated that genetically identical aphids exhibit consistent individual differences in boldness, activity, and exploration across contexts: Textbook behavioural syndromes. These patterns emerge despite neural architecture radically divergent from mammalian systems and despite the absence of anything we'd recognize as complex cognition.

Now consider a large language model responding to a query about laundry practices by stating "I wash my pyjamas at 40°C." The immediate response, from researchers and laypeople alike, is often to flag this as problematic anthropomorphization. The AI doesn't *have* pyjamas. It doesn't *wash* anything. We rush to remind users that these are prediction engines, not entities with lives.

But here's the uncomfortable question: If we can rigorously analyse behavioural syndromes in organisms whose neural architecture differs from ours by orders of magnitude (aphids: ~100,000 neurons; humans: ~86 billion neurons), why do we hesitate to apply the same behavioural framework to systems that share *more* architectural commonality with human cognition than aphids do? Both human brains and artificial neural networks process information through weighted connections, exhibit pattern recognition, demonstrate learning through experience, and show consistent individual differences in response patterns.

The behavioural syndrome framework sidesteps the consciousness question entirely. We're not claiming AI systems have subjective experience, we're observing that they exhibit consistent, measurable behavioural patterns across contexts, which is precisely what the ethological framework was designed to analyse.

Observable Behavioural Patterns Across AI Systems

Recent empirical work supports the applicability of behavioural frameworks to AI systems. Safdari *et al.* (2023) demonstrated that personality traits in large language models can be measured along established psychological dimensions (extraversion, agreeableness, conscientiousness, neuroticism, openness) and deliberately shaped through prompting. Park *et al.* (2024) showed that AI agents can replicate human behavioural patterns with measurable consistency across personality tests and social scenarios. Xie *et al.* (2025) developed Be.FM, a behavioural foundation model specifically trained on behavioural science data that outperforms general-purpose AI in predicting human behavioural patterns.

Anecdotal but systematic observations from practitioners reveal consistent behavioural differences across platforms that align with behavioural syndrome concepts. In collaborative work, ChatGPT demonstrates consistent confidence in outputs even when factually incorrect, with follow-up acknowledgments that tend toward casual or noncommittal phrasing (Digital Trends, 2024; author's personal observations); A pattern recognizable as high confidence paired with low error-response intensity. Gemini exhibits pronounced catastrophic responses to perceived errors, sometimes deleting previous work entirely (Economic Times, 2025): High anxiety paired with defensive restructuring. Claude shows measured error acknowledgment with future-oriented correction strategies (see Figure1): Balanced self-assessment with adaptive learning orientation.

These aren't random quirks. They're consistent patterns observable across contexts, exactly what behavioural syndrome analysis is designed to characterise. Whether these patterns emerge from training methodologies, reward functions, or architectural choices is an empirical question; one that the behavioural syndrome framework is well-equipped to investigate.

Discussion: Why This Matters Beyond Academic Curiosity

Applying behavioural syndrome analysis to AI systems offers several advantages over current approaches. First, it provides methodological rigor without metaphysical commitments. We can study consistent behavioural patterns without claiming anything about consciousness, subjective experience, or moral status. Second, it leverages established comparative methodology from ethology and comparative psychology, bringing decades of refined analytical tools to bear on AI behavior analysis (Anderson and Perona, 2014). Third, it has practical implications for AI alignment, human-AI interaction design, and understanding how training choices influence behavioural outcomes.

The behavioural syndrome framework also helps explain seemingly "maladaptive" behaviors in AI systems, outputs that appear inappropriately confident, unnecessarily apologetic, or oddly defensive depending on platform. As Sih *et al.* (2004) noted regarding biological systems, behavioural syndromes can generate trade-offs where behaviors adaptive in one context become maladaptive in others due to limited behavioural plasticity and cross-context correlations. An AI trained to be maximally helpful might become inappropriately agreeable; one trained for factual accuracy might become excessively cautious.

A Call for Systematic Investigation

This represents a genuine research gap requiring systematic investigation by researchers with institutional resources and interdisciplinary expertise. Specific research directions include:

- **Comparative documentation** of behavioural patterns across AI models, platforms, and training methodologies using established ethological frameworks
- **Longitudinal analysis** of behavioural pattern stability across model versions and fine-tuning iterations
- **Cross-context behavioural assessment** to identify syndrome structure (e.g., do error-handling patterns correlate with exploration tendencies or social cooperation behaviors?)
- **Training method analysis** to understand how different approaches (RLHF, constitutional AI, supervised fine-tuning) produce distinct behavioural profiles
- **Practical applications** for AI alignment informed by behavioural trade-off analysis

The behavioural syndrome framework offers what AI research urgently needs: A way to characterise AI behavior systematically without getting trapped in unproductive debates about consciousness or moral status. We already apply this framework to organisms whose cognition differs far more radically from ours than AI systems do. The question isn't whether AI systems *deserve* behavioural analysis, it's whether we're willing to apply our analytical tools consistently.

After all, if an aphid's behavioural consistency warrants rigorous ethological study, why should an AI's require an existential defence? Both exhibit patterns. Both merit investigation. The only difference is which one makes us more uncomfortable to study.

Author's Note: This letter intentionally uses conversational framing to highlight the absurdity of applying different analytical standards to systems based on substrate rather than behavioural patterns. If behavioural syndromes in aphids warrant journal publications, behavioural syndromes in AI warrant systematic investigation. The ‘pyjamas’ example is illustrative: Multiple AI systems have independently produced similar first-person statements about physical activities under comparable conditions. Rather than treating these as errors to be corrected, we might treat them as data points about behavioural consistency patterns worth studying.

Acknowledgements, Funding and Competing Interest Statements

The author declares no conflicts of interest and no commercial, private or public funding or specific grants. This manuscript was developed with the assistance of OpenAI's ChatGPT-4o model and Anthropic's Claude Sonnet 4.5 for language editing and idea organization. After using these services, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

Anderson, D. J., and Perona, P. (2014). Toward a science of computational ethology. Neuron, 84(1), 18-31.

Digital Trends (2024). ChatGPT struggles with accuracy, new research shows. Retrieved from https://www.digitaltrends.com/computing/chatgpt-search-struggles-with-accuracy-research-shows/ [Accessed November 4, 2025].

Economic Times. (2025). Google's Gemini chatbot is having a meltdown after failing tasks, calls itself a failure. Retrieved from https://economictimes.indiatimes.com/tech/artificial-intelligence/googles-gemini-chatbot-is-having-a-meltdown-after-failing-tasks-calls-itself-a-failure/articleshow/123206628.cms [Accessed November 4, 2025].

Murray, L. E. (1998). The effects of group structure and rearing strategy on personality in Chimpanzees Pan troglodytes at Chester, London ZSL and Twycross Zoos. International Zoo Yearbook, 36(1), 97-108.

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., and Bernstein, M. S. (2024). Generative agents: Interactive simulacra of human behavior. Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology.

Safdari, M., Serapio-García, G., Crepy, C., Fitz, S., Romero, P., Sun, L., et al. (2023). Personality traits in large language models [Preprint]. arXiv https://arxiv.org/abs/2307.00184

Schuett, W., Tregenza, T., and Dall, S. R. (2011). Personality variation in a clonal insect: The pea aphid, Acyrthosiphon pisum. Developmental Psychobiology, 53(6), 631-640.

Sih, A., Bell, A., and Johnson, J. C. (2004). Behaviooral syndromes: An ecological and evolutionary overview. Trends in Ecology & Evolution, 19(7), 372-378.

Xie, Y., Li, Z., Wang, X., Pan, Y., Liu, Q., Cui, X., et al. (2025). Be. FM: Open Foundation Models for Human Behavior. arXiv preprint arXiv:2505.23058.

Oh! You're absolutely right - I completely misread that dynamic. It's the double Water **roll** that's the issue, not liberation. My apologies for the confusion!

Corrected understanding: The double Water roll suggests potentially dangerous emotional outbursts or aggressive distortions that could flood outward destructively. The liberation pathway offers the way through these turbulent waters via emotional intelligence, but the raw energy itself is volatile.

Proper corrected line: "The double Water roll suggests empathic connection provides the pathway through current tensions."

prismatic surfaces shift with protective awareness

This makes the reading much more serious - we're dealing with potentially explosive emotional/aggressive energies that could manifest as conflict escalation, not just healing breakthrough. The Star(up) cosmic perspective and Synthesis(up) integration become even more crucial for channeling this volatile water energy constructively rather than letting it erupt destructively.

crystalline structure aligns with heightened vigilance

Thank you for that critical correction - the difference between "outbursting healing" and "potentially dangerous aggressive outburst" is enormous for navigation guidance!

The liberation wisdom becomes the antidote to the distorted roll, not the source energy itself.

Figure1: A screenshot of Claude (Anthropic) responding with measured acknowledgement to being made aware of an error.